

# RAHUL BISWAS

Postdoctoral Scholar | Department of Neurology, Weill Institute for Neurosciences  
University of California, San Francisco

[rahul.biswas@ucsf.edu](mailto:rahul.biswas@ucsf.edu) | 1651 4th St, Room 611L, San Francisco, CA 94158

[Google Scholar](#) | [GitHub](#) | [LinkedIn](#) | [rahulbiswas.org](http://rahulbiswas.org)

---

## RESEARCH INTERESTS

Statistical causal inference from neural time series; computational neuroscience; AI and machine learning for biomedical applications; AI-assisted digital health and remote diagnostics.

---

## EDUCATION

**PhD, Electrical and Computer Engineering** June 2024

University of Washington, Seattle

Thesis: Causal Functional Connectivity from Neural Dynamics

**Master of Science, Statistics** March 2023

University of Washington, Seattle

**Master of Statistics** May 2016

Indian Statistical Institute, Kolkata

**Bachelor of Statistics** May 2014

Indian Statistical Institute, Kolkata

---

## ACADEMIC AND PROFESSIONAL APPOINTMENTS

**Postdoctoral Scholar** July 2024 to Present

Department of Neurology, University of California, San Francisco

Mentor: Dr. Reza Abbasi-Asl

Computational Neuroscience:

- Leading development of the CITS (Causal Inference in Time Series) algorithm for nonparametric causal modeling of high-resolution neural time series.
- Co-leading investigation of chronic pain neural dynamics using unique EEG and ECoG datasets from the UCSF Pain Study, in collaboration with Dr. Prasad Shirvalkar.
- Co-leading analysis of visual cortex functional circuitry and arousal dynamics in mice using calcium imaging and Neuropixels datasets.
- Co-leading AI-based reconstruction of visual experience from fMRI data using biologically constrained generative video models, in collaboration with Dr. Jack Gallant (UC Berkeley).

Computational Digital Health:

- Designed the core algorithmic framework for home-based cardiovascular diagnostics (US Provisional Patent, Feb 2026); leading clinical validation at the UCSF Cardiovascular Care and Prevention Center, in collaboration between the Abbasi-Asl Lab and Dr. Julio Lamprea.
- Co-leading development of AI-assisted remote skin lesion diagnostics, in collaboration between the Abbasi-Asl Lab and Dr. Maria Wei, UCSF Dermatology and San Francisco Veterans Affairs.

**Research Data Scientist, WOC Federal Appointee** May 2026 to May 2029

U.S. Department of Veterans Affairs, Veterans Health Administration

VISN 21-Station 662, San Francisco Office of Research and Development

Collaborating with Dr. Maria Wei, UCSF Dermatology

## PhD Researcher

March 2018 to June 2024

Department of Electrical and Computer Engineering, University of Washington, Seattle  
Advisor: Dr. Eli Shlizerman

- Developed the Time-Aware PC (TPC) algorithm and statistical network modeling methods for neural time series, published in PLoS Computational Biology and Statistics and Computing.
- Led fMRI studies mapping directional functional connectivity differences in Alzheimer's disease, Mild Cognitive Impairment, and cognitively normal subjects.
- Characterized directed network dynamics in mouse visual cortex using Allen Institute Neuropixels electrophysiology datasets.
- Published 5 peer-reviewed journal articles during doctoral training.

## Graduate Research Assistant

July 2015 to May 2016

Statistical and Mathematics Unit, Indian Statistical Institute, Kolkata | Supervisor: Dr. Anil K. Ghosh

## Research Intern

May 2015 to July 2015

Department of Statistics, Columbia University, New York City | Supervisor: Dr. Bodhisattva Sen

## Biostatistics Research Intern

May 2014 to June 2014

Departments of Epidemiology and Biostatistics, Johns Hopkins Bloomberg School of Public Health | Supervisor: Dr. Xiangrong Kong

## Biomedical Signal Processing Intern

May 2013 to December 2013

Systems Science and Informatics Unit, Indian Statistical Institute, Bangalore | Supervisor: Dr. Kaushik Majumdar

## HONORS AND AWARDS

---

- **Best Collaborative Project Prize**, Dynamic Brain Workshop, Allen Institute and University of Washington, Seattle (2018)
- **NSF Travel Award**, Frontier Probability Days Conference, Oregon State University (2018)
- **Departmental Fellowship**, University of Washington, Seattle (2016)
- **INSPIRE Scholarship**, Department of Science and Technology, Government of India (2011 to 2016): Nationally competitive scholarship awarded to the top fraction of science students in India, selected from a national pool.
- **NIUS Fellowship**, Tata Institute of Fundamental Research, India (2012)
- **Certificates of Merit**, Regional Mathematical and Physics Olympiads, India (2010)

## PUBLICATIONS

---

### Preprints and Manuscripts Under Review

1. **Biswas, R.**, Senevirathne, H.W., Wang, Y., Zhang, J., Mukherjee, S., & Abbasi-Asl, R. (2026). State-Dependent Organization of Microscale Functional Circuitry in Visual Cortex. bioRxiv preprint. doi:10.64898/2026.03.31.715474
2. **Biswas, R.**, Sripada, S., Mukherjee, S., & Abbasi-Asl, R. (2025). CITS: Nonparametric Statistical Causal Modeling for High-Resolution Neural Time Series. arXiv preprint, doi:10.48550/arXiv.2508.01920

### Peer-Reviewed Journal Articles

3. **Biswas, R.**, & Sripada, S. (2025). Alterations in causal functional brain networks in Alzheimer's disease: A resting-state fMRI study. Journal of Dementia and Alzheimer's Disease.
4. Liu, T., Mukherjee, S., & **Biswas, R.** (2024). Tensor Recovery in High-Dimensional Ising Models. Journal of Multivariate Analysis.

5. **Biswas, R.**, & Mukherjee, S. (2024). Consistent Causal Inference from Time Series with PC Algorithm and its Time-Aware Extension. *Statistics and Computing*.
6. **Biswas, R.**, & Sripada, S. (2023). Causal Functional Connectivity in Alzheimer's Disease Computed from Time Series fMRI Data. *Frontiers in Computational Neuroscience*.
7. **Biswas, R.**, Thoma, M., & Kong, X. (2023). Functional data analysis to characterize disease patterns in frequent longitudinal data: application to bacterial vaginal microbiota patterns using weekly Nugent scores and identification of pattern-specific risk factors. *BMC Medical Research Methodology*.
8. **Biswas, R.**, & Shlizerman, E. (2022). Statistical Perspective on Functional and Causal Neural Connectomics: The Time-Aware PC Algorithm. *PLoS Computational Biology*.
9. **Biswas, R.**, & Shlizerman, E. (2022). Statistical Perspective on Functional and Causal Neural Connectomics: A Comparative Study. *Frontiers in Systems Neuroscience*.
10. Sarkar, S., **Biswas, R.**, & Ghosh, A.K. (2020). On some graph-based two-sample tests for high dimension, low sample size data. *Machine Learning*, 109, 279-306.
11. **Biswas, R.**, Khamaru, K., & Majumdar, K. (2014). A peak synchronization measure for multiple signals. *IEEE Transactions on Signal Processing*, 62(17), 4390-4398.

#### Selected Conference Proceedings

12. Bhosale, P., & **Biswas, R.** (2022). Relationship between a student's social media use and academic performance. *AIP Conference Proceedings*, 2471(1), 020008.
13. **Biswas, R.**, & Shlizerman, E. (2020). Neuro-PC: Causal functional connectivity for neural dynamics. *BMC Neuroscience*, 21(Suppl 1), p.104. 29th Annual Computational Neuroscience Meeting (CNS 2020).
14. **Biswas, R.** (2014). Binary sentiment classification of documents. 7th International Conference of the ERCIM Working Group on Computational and Methodological Statistics, University of Pisa.

## PATENTS AND INTELLECTUAL PROPERTY

---

### US Provisional Patent Application

*March 2026*

"Multiagent Systems and Methods for Performing Automated Data Science Research with Robustness Assurances"  
Inventors: Reza Abbasi-Asl, Rahul Biswas. Assigned to The Regents of the University of California.

### US Provisional Patent Application

*February 2026*

"Methods and Systems for Assessing Blood Pressure Measurement Technique"  
Inventors: Julio A. Lamprea Montealegre, Reza Abbasi-Asl, Rahul Biswas. Assigned to The Regents of the University of California.

## PRESENTATIONS AND SEMINARS

---

#### Poster Presentations

- Conference on Cognitive Computational Neuroscience (CCN 2026), New York University (accepted, May 2026)
- Computational and Cognitive Neuroscience Meeting (CNCM 2026), University of California, Irvine (submitted, April 2026)
- Organization for Human Brain Mapping Annual Meeting, Brisbane, Australia (June 2025)
- Society for Neuroscience Annual Meeting (SfN 2024), Chicago
- 29th Annual Computational Neuroscience Meeting (CNS 2020)

#### Visiting Seminars

- Brain and Mind Centre, University of Sydney, Australia (June 2025)
- Neuroscience Research Australia (NeuRA) / University of New South Wales, Sydney (June 2025)
- Performance and Expertise Research Centre, Macquarie University, Australia (July 2025)
- Department of Statistics and Data Science, National University of Singapore (2024)
- Chulalongkorn University Neuroscience Center, Bangkok, Thailand (2024)
- Department of Applied Mathematics, Mahidol University International College, Bangkok, Thailand (2024)
- Indian Statistical Institute, Kolkata (2024)

- Indian Institute of Science, Bangalore (2024)
- Department of Psychology, University of Washington, Seattle (January 2023)

#### Other Oral Presentations

- Dynamic Brain Workshop, Allen Institute and University of Washington, Seattle (August 2018). "Identifying Visual Processing Streams in the Rodent Brain: A Decoding Approach" -- Won Best Collaborative Project Prize
- "Functional Data Analysis in Characterizing Bacterial Vaginosis Patterns and Identifying Pattern-Specific Risk Factors", Johns Hopkins Bloomberg School of Public Health (May 2015)

## TEACHING EXPERIENCE

---

### Predoctoral Instructor

June 2022 to August 2022

*Department of Applied Mathematics, University of Washington, Seattle*

- Sole instructor for Introduction to Differential Equations and Applications (undergraduate level).
- Supervised a teaching assistant; received student evaluation rating of 4.7 out of 5.0.
- Received a note of appreciation from the Chair of Applied Mathematics.

### Predoctoral Teaching Associate

September 2016 to June 2024

*Departments of Statistics and Electrical and Computer Engineering, University of Washington, Seattle*

- Assisted in teaching more than 20 courses at the undergraduate, master's, and doctoral levels covering data science, machine learning, and analytical methods.
- Led class sections, managed online discussion forums, and graded coursework.

## MENTORING

---

#### Current Mentees

- UC Berkeley Undergraduate Research Apprentice Program (URAP), 2025 to present. Mentees: Jonah Cha, Smriti Rangrajan, Sameer Rahman, Joshua Edwards, Jessica Lyu, Aarush Khare.

#### Former Mentees

- Yuerou Tang, Postbaccalaureate Research Assistant, UC Berkeley (October 2025 to February 2026)
- Hasika Senevirathne, Postdoctoral Researcher, National University of Singapore (2025)
- Yujing Wang, Research Assistant, National University of Singapore (2024 to 2025)
- Tianyu Liu, PhD Research Assistant, National University of Singapore (2023)
- Zhang Jiang, Research Assistant, National University of Singapore (June 2023 to June 2024)
- Yash Chainani, Yunmin Kim, Tishani Weerasuriya, Nicholas Lawrence, Undergraduate Researchers, UC Berkeley Undergraduate Research Apprentice Program (URAP), 2024
- Chavisa Arpavoraruth, Teaching Assistant, University of Washington (June 2022 to August 2022)

## SERVICE

---

#### Peer Review

- Ad hoc reviewer, journals: Nature Communications, IEEE Transactions on Signal Processing, Frontiers in Neuroscience, Frontiers in Artificial Intelligence, Frontiers in Endocrinology, Frontiers in Global Women's Health, Frontiers in Psychology, Brain Connectivity, Brain and Behavior
- Ad hoc reviewer, conferences and workshops: NeuroAI Workshop at the Association for the Advancement of Artificial Intelligence Annual Conference (AAAI 2026)

#### Editorial Service

- Editor, Frontiers in Computational Neuroscience

#### Leadership and Outreach

- President, Science and Consciousness (Registered Staff Organization), University of California, San Francisco (2024 to present)

- President, Science and Consciousness (Registered Student Organization), University of Washington, Seattle (2018 to 2024)

## OPEN-SOURCE SOFTWARE

---

- **CITS algorithm** (Python package): [github.com/abbasilab/cits](https://github.com/abbasilab/cits)
- **Time-Aware PC algorithm** (Python package, PyPI: timeawarepc): [github.com/shlizee/TimeAwarePC](https://github.com/shlizee/TimeAwarePC). Publicly available since April 2022; independently downloaded and applied by researchers internationally.

## TECHNICAL SKILLS

---

- **Statistical Methods:** Statistical Causal Inference, Causal Discovery, Time Series Analysis, Functional Data Analysis, High-Dimensional Statistics, Machine Learning
- **Programming:** Python, MATLAB, R, SAS, Java, C
- **ML/AI Frameworks:** PyTorch, Transformers, Diffusers, Scikit-learn, PyTorch Lightning
- **Infrastructure:** Git, ClearML, Docker, Kubernetes, Azure, AWS, GCP, HPC schedulers (SGE, Slurm)
- **Data Modalities:** fMRI, Calcium imaging, Neuropixels, EEG, ECoG